

Md. Tahmid Ashraf Chowdhury | Graduate Assistant & MSc Researcher

Location: Seri Iskandar, Perak, Malaysia

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EDUCATION

Jan 2025 – Present	Master of Science (MSc) in Information Technology - Research Track <i>Universiti Teknologi PETRONAS (UTP), Perak, Malaysia - Department of Computing</i> Thesis area: Blockchain-based fractional real-estate tokenization using Polygon Layer 2, ERC-1155 smart contracts, and secure API integration; broader research across federated learning, Byzantine-resilient distributed AI, blockchain auditability, and IoT/critical-infrastructure cybersecurity. Milestone: Awarded a Graduate Assistantship (theory lectures and laboratory instruction).
2020 – 2023	Bachelor of Science (BSc) in Computer Science & Engineering (Major: Software Engineering) <i>American International University-Bangladesh (AIUB)</i> Milestone: <i>Magna Cum Laude</i> , CGPA 3.81/4.00; Dean's Award for Academic Excellence. Medium of instruction: English.

RESEARCH INTERESTS

• Federated & distributed learning • Byzantine fault tolerance & robust aggregation • Blockchain-based auditability & decentralised trust • Smart contracts for accountable AI • Privacy-preserving & trustworthy AI • Cybersecurity for IoT/IIoT & cyber-physical systems • AI/ML for anomaly & intrusion detection • Edge intelligence & critical-infrastructure security

HONORS AND AWARDS

2025	Graduate Assistantship Awarded by Universiti Teknologi PETRONAS (UTP), Perak, Malaysia.
2023	Dean's Award for Academic Excellence Awarded by American International University-Bangladesh (AIUB).
2023	<i>Magna Cum Laude</i> (CGPA 3.81/4.00) Awarded by American International University-Bangladesh (AIUB).
2021	National Programming Contest National-level programming competition, Bangladesh, 2021.

LANGUAGE PROFICIENCY

- **English - IELTS Academic:** Overall 6.5 (Listening 7.0, Reading 7.0, Writing 6.0, Speaking 6.0; CEFR B2), British Council, March 2024.

PUBLICATIONS

Selected Journal Articles

1. **M. T. A. Chowdhury**, F. Ullah, et al. "LLM-Guided Population-Based Reinforcement Learning: A Scalable Methodology for Adaptive Hyperparameter Optimization." *MethodsX*, vol. 16, 103879, 2026. [\[GitHub\]](#)
2. **M. T. A. Chowdhury**, et al. "RollupFL: An Auditable Federated Learning Framework for Byzantine Client Accountability." *International Journal of Advanced Computer Science and Applications (IJACSA)*, 2026. [\[GitHub\]](#)
3. **M. T. A. Chowdhury**, et al. "SAB-Select: Staleness-Aware Burst-Adaptive Client Selection for Federated Learning under Bursty Connectivity." *International Journal of Mathematical, Engineering and Management Sciences (IJMEMS)*, 2026. [\[GitHub\]](#)
4. **M. T. A. Chowdhury**, et al. "Predicting Learning Outcomes and Engagement from AR/VR Education Data: A Cross-Dataset Machine-Learning Study." *International Journal of Technology in*

Selected Conference Proceedings

1. **M. T. A. Chowdhury**, et al. "A Blockchain-Based Conceptual Framework for Real-Time Collision Prevention in Autonomous Vehicles." *IEEE Int. Conf. on Software Engineering & Computer Systems (ICSECS)*, 2025. [IEEE Xplore](#)
2. **M. T. A. Chowdhury**, et al. "Blockchain-Enabled Federated Learning for Collaborative Autonomous Vehicle Navigation." *ICMCTC*, 2025. [IEEE Xplore](#)
3. **M. T. A. Chowdhury**, et al. "Data Quality Assessment of Distributed Acoustic Sensing for CO₂ Leak Detection Using Machine Learning." *AiDAS*, 2025. [IEEE Xplore](#)
4. **M. T. A. Chowdhury**, et al. "FedFOMAML: Federated First-Order Meta-Learning for Next-Beam Prediction in mmWave Networks." *Multimedia University Engineering Conference (MECON)*, 2026 (Accepted).

Books and Chapters

1. **M. T. A. Chowdhury**, et al. "Blockchain Synergies with the Internet of Everything (IoE)." In *Convergence of Blockchain, IoE and Federated Learning for Security*, [IGI Global](#), 2025.
2. **M. T. A. Chowdhury**, et al. "Leveraging Smart Contracts for Enhanced IoE Security." [IGI Global](#), 2025.
3. **M. T. A. Chowdhury**, et al. "Blockchain and IoE Transforming Industry Applications." [IGI Global](#), 2025.
4. **M. T. A. Chowdhury**, et al. "Enhancing Transparency and Efficiency: Blockchain-Based Supply Chain Management and Automated Trade Finance in Oil and Gas." [IGI Global](#), 2026.

Manuscripts Under Review

1. **M. T. A. Chowdhury**, et al. "Reliability-Certified Byzantine-Resilient Federated Meta-Learning for Auditable Remaining Useful Life Prognostics." *IEEE Transactions on Reliability*. [\[GitHub\]](#)
2. **M. T. A. Chowdhury**, et al. "Bio-Chain: A DAG-Based Trust Framework for Secure Molecular Event Validation and Actuation in the Internet of Bio-Nano Things." *Scientific Reports*. [\[GitHub\]](#)
3. **M. T. A. Chowdhury**, et al. "LLM-Assisted Reinforcement Learning Framework for AR and VR Educational Environments." *Scientific Reports*. [\[GitHub\]](#)
4. **M. T. A. Chowdhury**, et al. "Blockchain-Based Real-Estate Tokenization as a Big-Data and Distributed-Ledger Ecosystem: A Systematic Review of Applications, Data Challenges and Future Directions." *Journal of Big Data*. [\[GitHub\]](#)
5. **M. T. A. Chowdhury**, et al. "Federated Learning for Climate-Resilient Smart Buildings: A Privacy-Preserving Approach." *Int. Conf. on Regenerative Futures*, 2026.

JOB EXPERIENCE

Jan 2025 – Present	Graduate Assistant / MSc Researcher <i>Universiti Teknologi PETRONAS (UTP) - Seri Iskandar, Malaysia</i> • Developed RollupFL, a blockchain-anchored federated-learning system providing auditable accountability for distributed nodes under Byzantine attack, decoupling robust aggregation from an integrity/attribution audit layer. • Designing and prototyping a scalable blockchain tokenization architecture for secure, decentralised data sharing across distributed asset networks. • Authored three IGI Global book chapters on blockchain-IoE integration, smart-contract security, and industry deployment of blockchain architectures.
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May 2023 – Jan
2025

Software Engineer

GIGA Tech Limited - Dhaka, Bangladesh

- Architected and delivered government-scale distributed data-management systems (Java Spring Boot microservices, REST and GraphQL APIs) serving national food-ministry operations.
- Engineered a real-time OCR-powered identity-verification module; led code reviews, performance optimisation, and mentoring of junior engineers.

COMPUTING SKILLS

- **Distributed / Federated ML:** Federated learning, Byzantine-robust aggregation (Krum, Median, trimmed mean), federated meta-learning, PyTorch, scalable system design.
- **Blockchain & Security:** Solidity smart contracts, Ethereum / Polygon L2, ERC-20/1155/1400, Web3, consensus design, decentralised trust and auditability.
- **AI & ML:** Deep learning, reinforcement learning, LLM integration, anomaly / intrusion detection, AI-agent design.
- **Programming & Backend:** Python, Java, JavaScript/TypeScript, Solidity; Spring Boot, GraphQL, REST, Hibernate, JWT.
- **Tools & DevOps:** Git/GitHub/GitLab, Docker, Jest, React Testing Library.

LEADERSHIP & PROFESSIONAL SERVICE

- Vice-President, Post-Graduate Student Council - Department of Computing, UTP.
- Campus Ambassador - International Model United Nations (AIUB).
- Organiser - The HunGRE Games / Students for Humanity (AIUB).

REFERENCES

Assoc. Prof. Ts. Dr. Fasee Ullah

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